Right Triangle Trigonometry (4.5) Name ____________________

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write "2" in the blank. Solve that problem, find your answer and write "3" in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $\sqrt{3}$ #______ In a right triangle, the side opposite θ is 3 and the hypotenuse is 5. Find $\sin\theta$.	ANSWER: $\frac{8}{17}$ #______ In a right triangle, opposite = 8 and adjacent = 15. Find $\tan\theta$.
ANSWER: $\frac{1}{2}$ #______ Evaluate $\sin 45°$.	ANSWER: $\frac{\sqrt{3}}{2}$ #______ Using a cofunction identity, find the angle whose sine equals $\cos 70°$.

Answer: $\dfrac{3}{5}$

________ Evaluate $\cos 60°$.

Answer: $\dfrac{24}{25}$

________ Evaluate $\cos 30°$.

Answer: $\dfrac{\sqrt{3}}{3}$

________ In a right triangle, the side adjacent to θ is 8 and the hypotenuse is 17. Find $\cos\theta$.

Answer: $20°$

________ Evaluate $\tan 60°$.

Answer: $\dfrac{\sqrt{2}}{2}$

________ Evaluate $\tan 30°$.

Answer: $\dfrac{8}{15}$

________ If θ is acute and $\sin\theta = \dfrac{7}{25}$, find $\cos\theta$.